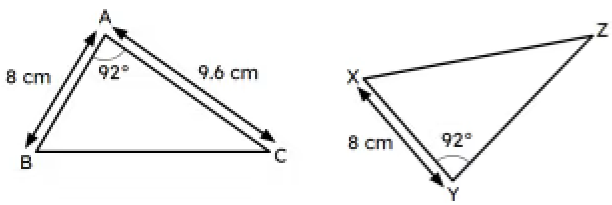
Answers

Exit quiz

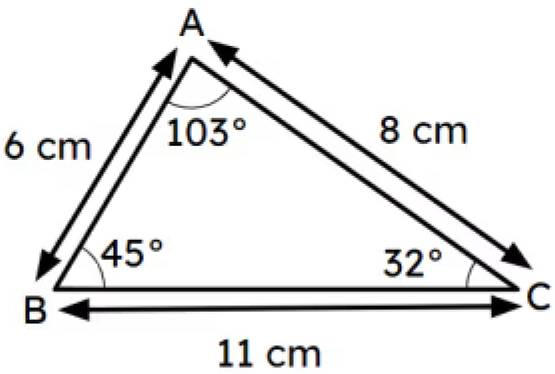
Congruent triangles (SAS)

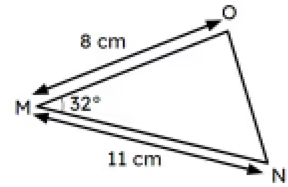
1 When there are two triangles, if the two corresponding edges and the angle between them are the same, you can prove they are congruent by SAS (side-angle-side). Fill in the blank

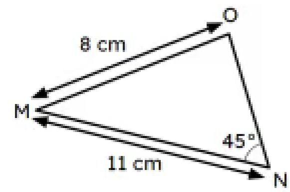
2 If triangles ABC and XYZ are congruent by SAS then YZ is 9.6 cm. Fill in the blank

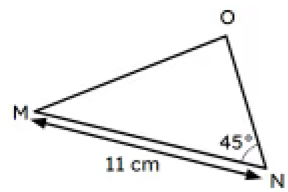


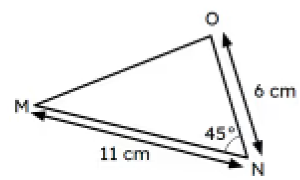
3 Which of the following triangles have sufficient information to guarantee they are congruent to triangle ABC? Tick **2** correct answers



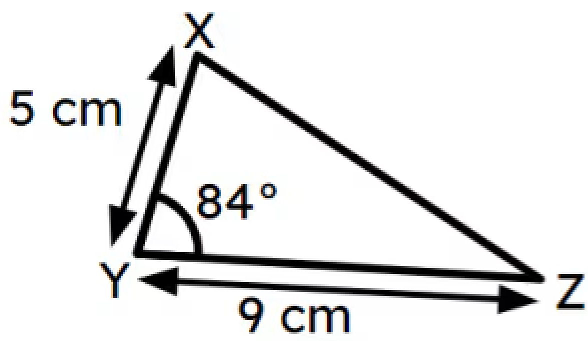
- ✓ 

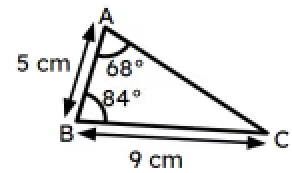
- ☐ 

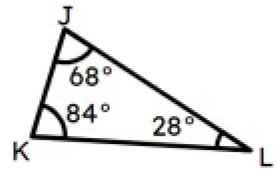
- ☐ 

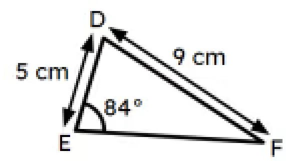
- ✓ 

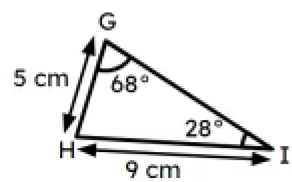
4 Which of the following triangles are congruent to triangle XYZ? Tick **2** correct answers



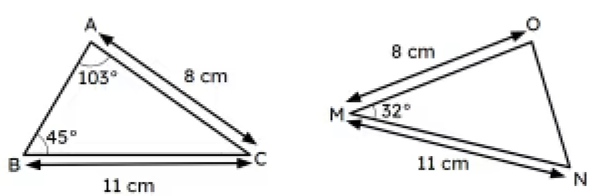
✓ 

☐ 

☐ 

✓ 

5 Are the triangles ABC and MNO congruent? Tick **1** correct answer

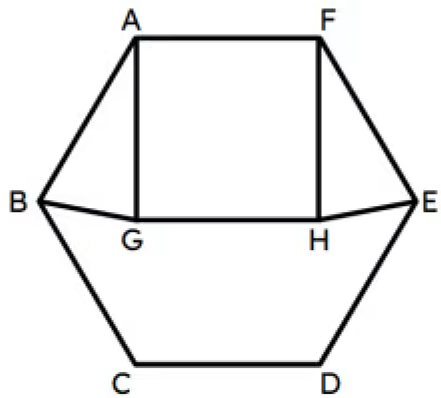


✓ Yes, they are guaranteed to be congruent.

☐ No, they are definitely incongruent.

☐ There is not enough information to tell.

6 This diagram shows a regular hexagon ABCDEF and a square AGHF. Match the statements to prove triangle ABG and FHE are congruent by SAS. Write the correct letter in each box



a	∠BAF = ∠EFA
b	∠GAF = ∠HFA
c	∠BAG =
d	∠HFE =
e	∠BAG =
f	AB = AG = FH = FE

a	as they are interior angles of a regular hexagon (= 120°)
d	∠EFA − ∠HFA (= 30°)
f	as they are in regular polygons with shared edges
b	as they are interior angles of a square (= 90°)
c	∠BAF − ∠GAF (= 30°)
e	∠HFE